

Recipe Food ERP

Software Test Report — Full System (business logic, configuration, build)

Recipe Food Products Limited · Test date: 2 October 2026 · Build under test: main @ 2f390c7 ("final version")

117 Test cases executed	117 Passed	0 Failed	100.0% Pass rate
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1. Executive summary

117 test cases were run against the Recipe Food ERP: 115 automated business-logic and configuration checks across 13 modules, plus a full TypeScript type check and a production build. 117 passed and 0 failed. The first run found one defect (DEF-01): the Dashboard counted **rejected** invoices in its "Top dealers" and "Top products" panels. It has been fixed, and every test passes on the re-run.

All money calculations behind Sales Reports, Company Earnings, Fund/Cash Flow, Vendor dues, Loans, Salary/Advances and Accounting (General Ledger, Trial Balance, Balance Sheet) returned the hand-calculated figures. Cross-checks that tie modules together also held: Company Earnings equals the category-report company profit, a vendor statement's current due equals the vendor due, a loan schedule's last closing equals the loan balance, September's closing cash carries into October's opening, and the product ledger's final balance equals current factory stock.

Verdict: all automated tests pass, and the only defect found (DEF-01) has been fixed and re-tested. Manual UI and end-to-end testing on a non-production Firebase project is still recommended (see Section 6).

2. Results by module

Module	Test cases	Passed	Failed	Status
Core Utilities	20	20	0	PASS
Dealer & Sale Type	6	6	0	PASS
Input & Authorization	9	9	0	PASS
Sales Reports	11	11	0	PASS
Company Earnings & P&L	7	7	0	PASS
Fund / Cash Flow	13	13	0	PASS
Purchase & Vendor	5	5	0	PASS
Factory Stock & Production	7	7	0	PASS
Loans, Salary & Advances	10	10	0	PASS
Accounting (GL / TB / BS)	7	7	0	PASS
Dashboard	4	4	0	PASS
Users, Roles & Permissions	5	5	0	PASS
Configuration & Data Integrity	11	11	0	PASS
Build & Compile	2	2	0	PASS
Total	117	117	0	

3. Defects found and fixed

ID	DEF-01
Title	Dashboard "Top dealers" and "Top products" include rejected invoices
Severity / Priority	Medium / High
Status	FIXED — re-tested 2 Oct 2026; TC-097 and TC-098 now pass
Failed tests (first run)	TC-097, TC-098
Location	src/lib/erp/utils.ts:206 — buildOperationsOverview() (used by DashboardScreen.tsx:52)
Steps to reproduce	1. Save an invoice for Dealer B worth 99,999. 2. Reject it on the Approvals page. 3. Open the Dashboard.
Expected	Rejected entries are not counted in money totals (the Input & Authorization rule). Dealer A (3,000) is the top dealer and Dealer B shows 300.
Actual (before fix)	Dealer B showed 100,299 and ranked first. "Turmeric 200g" showed 100,299 and ranked first in Top products.
Root cause	buildOperationsOverview() reads every rate card without the isCountedEntry() filter. Every other report applies that filter, including Sales Reports, Earnings and Cash Flow.
Fix applied	utils.ts buildOperationsOverview(): const rateCards = sortByCreatedAtDesc(toArray(data?.rateCards).filter(isCountedEntry))

Observation (not counted as a failure — needs a business decision)

OBS-01 — Rejected purchases still appear in Factory Stock and the Product Statement. computeFactoryStock() and computeProductLedger() (utils.ts) do not filter purchases by approval status. For stock quantities this matches the documented rule that "stock it moved stays moved until the entry is edited or deleted". However, the Product Statement's vendor-wise amount (totalPurchaseAmount) is a money figure, and it will include a rejected purchase. Please confirm whether the Product Statement should hide rejected purchases.

4. Scope and approach

In scope. All calculation and rule logic the ERP screens depend on, in src/lib/erp (utils.ts, approvals.ts, defaultData.ts, standardChartOfAccounts.ts, companyInfo.ts, starterCatalog.ts). This logic produces every total, report, due, balance and permission decision shown in the UI. Each test feeds an in-memory dataset with known values, including rejected and pending entries, legacy records, Bangla digits, date-range edges and year boundaries. The output is compared with figures calculated by hand. Configuration integrity checks cover the Firebase security rules, sidebar routes, permission catalogue and chart of accounts. The whole project was also type-checked and given a production build.

Not in scope. Click-through UI testing in a browser, and tests that write to Firebase. The development app is connected to the client's **live** database, so no live data was created, changed or deleted during this run. Firebase security rules were checked for coverage only; they were not run in the Firebase emulator. PDF/Excel export rendering was not tested because it needs a browser.

Test data. Sales fixture: 4 invoices (1 rejected), 3 collections (1 legacy, 1 rejected), 3 product returns (dealer, depot, rejected), 4 expenses, 7 cash entries, 7 loan transactions, 3 purchases, 3 vendor payments and 3 orders. A separate factory-stock fixture covers purchases, sales, manual usage and production usage on both sides of a date range. A separate accounting fixture covers a chart of accounts with opening balances, balanced ledger postings and a legacy "customer" alias.

5. Test environment

Application	Recipe Food ERP (Next.js 16.2.10, React 19, Firebase Realtime Database)
Branch / commit	main @ 2f390c7 + DEF-01 fix (src/lib/erp/utils.ts, not yet committed)
Runtime	Node.js v22.22.1 on Linux; timezone Asia/Dhaka (also re-run in UTC with identical results)

Test runner	Node assert-based suite (erp.test.js) on modules compiled with TypeScript 5.9; no database access
Build checks	tsc --noEmit (project tsconfig); next build --webpack
Books start date	2026-09-01 (BOOKS_START_DATE)

6. Recommendations

- Commit the DEF-01 fix and deploy it with the next release.
- Decide on OBS-01: should the Product Statement's vendor amount exclude rejected purchases?
- Before go-live, do a manual UI walkthrough on a **separate test Firebase project**, not the live one. Cover these flows: Invoice → Approval → Collection; Product Return with due credit; Purchase → Vendor payment; Production entry; Cash entry and Advance salary; Loan; Journal voucher; and user roles.
- Run database.rules.json in the Firebase emulator with one user per role, to confirm that write permissions match the role matrix.
- Keep this suite as a regression check, and re-run it after every change to the report calculations.

Appendix — Test case details

Each test case lists the module, what was checked, the expected result and the actual result. Bangla category names are shown in English here (Salary, Damage).

Core Utilities — 20 test cases

ID	Test case	Expected result	Actual result	Status
TC-001	Per Ctn/Bgs multiplier — "24 pcs = 1 Ct"	24	24	PASS
TC-002	Per Ctn/Bgs multiplier — "500 ps = 1 bg"	500	500	PASS
TC-003	Per Ctn/Bgs multiplier — thousands separator "1,000 pcs"	1000	1000	PASS
TC-004	Per Ctn/Bgs multiplier — Bangla digits "24 pieces" (Bangla digits)	24	24	PASS
TC-005	Per Ctn/Bgs multiplier — blank / undefined / text / zero default to 1	1 for all four	1, 1, 1, 1	PASS
TC-006	Bangla → Latin digit conversion	Bangla digits 0-9 → "0123456789"	0123456789	PASS
TC-007	Discount product rate chain (Depot SR 100, SR +10%, TP +20%)	SR Rate 110, TP Rate 132	SR Rate 110.00, TP Rate 132.00	PASS
TC-008	Product stock status thresholds	0→out-of-stock, -1→out-of-stock, 5/5→low-stock, 6/5→active	As expected	PASS
TC-009	formatDate guards invalid / empty dates (no crash)	"—" for "" and "not-a-date"	—	PASS
TC-010	formatDate renders a valid ISO date	Contains year 2026	Sep 1, 2026	PASS
TC-011	formatCurrency BDT with thousands grouping	Contains "1,500"	BDT 1,500	PASS
TC-012	createId format and uniqueness (1,000 ids)	prefix_timestamp_suffix; 1000 unique	1000 unique ids	PASS
TC-013	Dhaka "today" crosses UTC midnight correctly	2026-10-01T20:00Z → 2026-10-02	2026-10-02	PASS
TC-014	Inclusive date-range filter	from/to inclusive, open ends allowed	As expected	PASS
TC-015	sortByCreatedAtDesc / toArray helpers	Newest first; null → []	As expected	PASS
TC-016	Pack weight conversion 45 g × 72 pcs	3.24 kg; description string	45 g × 72 pcs = 3,240 g = 3.24 kg	PASS
TC-017	Packaging material available pieces (float-safe)	2.01 kg ÷ 1 g = 2010; 3 ctn × 24 = 72; plain raw → undefined	2010 / 72 / undefined	PASS
TC-018	Rate Card line → stock units (with / without pack conversion)	24 pcs × 2 ÷ 48 = 1 unit; without FG = 48 pcs	1 / 48	PASS
TC-019	Expense category display label (rename)	Renamed → "House Rent"; blank label → stored key	As expected	PASS
TC-020	Strong password rule (provider.isStrongPassword)	≥8 chars with letter + digit	abc12345 ✓; others rejected	PASS

Dealer & Sale Type — 6 test cases

ID	Test case	Expected result	Actual result	Status
TC-021	Commission sale type detection	legacy "commission" & "Commission Distributor" → true; others/trade → false	As expected	PASS
TC-022	Trade Sales type detection	Only "Trade Sales" category → true	As expected	PASS
TC-023	SR distributor detection uses whole word "SR"	"SR Distributor" true; "Mister Distribution" false	As expected	PASS

ID	Test case	Expected result	Actual result	Status
TC-024	Sale type label resolution	Category name / legacy label / undefined	As expected	PASS
TC-025	Dealer due excludes cancelled orders	100 + 25 = 125	125	PASS
TC-026	Dealer lifetime totals from orders	d1 total = 550	550	PASS

Input & Authorization — 9 test cases

ID	Test case	Expected result	Actual result	Status
TC-027	Rejected entries are not counted; pending/approved/legacy are	rejected → false; pending/approved/none/null → true	As expected	PASS
TC-028	Missing approval status is treated as approved (legacy data)	"approved"	As expected	PASS
TC-029	New/edited entry fields reset to pending	status pending, submitter set, approver cleared	As expected	PASS
TC-030	Approval queue lists only records with an explicit status	3 items (legacy invoice skipped)	3	PASS
TC-031	Approval queue is newest-first by submission time	PU-C, INV-A, JV-1	As expected	PASS
TC-032	Approval rows carry department, permission and description	Invoice→sales/orders:approve; JV amount = Σdebit 150; purchase detail lists materials	As expected	PASS
TC-033	Approval queue on null data	[]	As expected	PASS
TC-034	Every approval permission exists in the permission catalogue	All APPROVAL_PERMISSION_IDS ⊆ ALL_PERMISSION_IDS	orders:approve, purchase:approve, finance:approve, accounting:approve	PASS
TC-035	Approval registry covers 11 department collections	11 collections, 5 departments	As expected	PASS

Sales Reports — 11 test cases

ID	Test case	Expected result	Actual result	Status
TC-036	Sales summary counts only non-rejected invoices	3 invoices, total 3,300	3 / 3300	PASS
TC-037	Dealer returns net off sales (depot & rejected returns excluded)	Returns 100, Net 3,200	As expected	PASS
TC-038	Sale-type split commission / others / unclassified	2,400 / 300 / 600	As expected	PASS
TC-039	Dealer-wise row for Dealer A	2 invoices, total 3,000, return 100, net 2,900	As expected	PASS
TC-040	Product-wise row applies per-carton & Bangla multipliers	Chili: 60 pcs, 3,000, net 2,900	As expected	PASS
TC-041	Date-wise rows include return-only dates	2026-09-25: total 0, return 100, net -100	As expected	PASS
TC-042	Sales summary scoped to September	2 invoices, 2,700; net 2,600	As expected	PASS
TC-043	Category-wise sales with returns netted	Spices dealer 2,900 / company profit 290 / depot profit 580	As expected	PASS
TC-044	Category totals	Amount 3,140; company profit 330; depot profit 620	As expected	PASS
TC-045	Line without product link → "Uncategorized"	Uncategorized row present	As expected	PASS
TC-046	Empty / null data returns zeroed reports	No crash, totals 0	As expected	PASS

Company Earnings & P&L — 7 test cases

ID	Test case	Expected result	Actual result	Status
TC-047	Total earning = Σ usableMoney – return company profit (rejected excluded)	$240+50+60 - (10+10) = 330$	330	PASS
TC-048	Earning equals category-report company profit (cross-check)	$330 = 330$	As expected	PASS
TC-049	Total expense = expenses + direct-expense cash (Advance/Direct), rejected excluded	$500+8000+3000+150+3000+1000 = 15,650$	15650	PASS
TC-050	Net profit	$330 - 15,650 = -15,320$	As expected	PASS
TC-051	Average profit ratio on dealer value sales	$330 \div 3,140 = 10.51 \%$	10.51 %	PASS
TC-052	Yearly breakdown sums to totals	One year (2026): earning 330, expense 15,650	As expected	PASS
TC-053	Monthly window has requested number of months	6 months	As expected	PASS

Fund / Cash Flow — 13 test cases

ID	Test case	Expected result	Actual result	Status
TC-054	September inflow split (sales / legacy / loans / other)	Sales 1,300; Legacy 700; Loans 10,000; Other 200; Total 12,200	As expected	PASS
TC-055	Opening loan balance & rejected loans are not cash in	Loans = 10,000 only	As expected	PASS
TC-056	September outflow (expenses + cash out, rejected excluded)	$8,500 + 5,150 = 13,650$	As expected	PASS
TC-057	Opening balance on books start date is zero	0	As expected	PASS
TC-058	Closing balance	$0 + 12,200 - 13,650 = -1,450$	As expected	PASS
TC-059	September closing carries into October opening	Oct opening –1,450; Oct closing –4,450	As expected	PASS
TC-060	Outflow by category merges Expense + Cash charts	Salary 8,000 top; categories total = outflow	As expected	PASS
TC-061	Range P&L	Earning 270; expense 12,650; net –12,380	As expected	PASS
TC-062	Product returns in range	2 returns, value 130, deducted 20	As expected	PASS
TC-063	Vendor-wise purchases (rejected excluded)	Vendor One: 2 purchases, 6,200, paid 3,500, due 2,700	As expected	PASS
TC-064	Item-wise purchased materials and products sold	Chili Raw 100 kg/5,000; Chili 100g 48 pcs/2,400	As expected	PASS
TC-065	Legacy collection detection	No rateCardId or unknown invoice → legacy	As expected	PASS
TC-066	Cash direction helpers ignore rejected rows	out/in/direct flags correct	As expected	PASS

Purchase & Vendor — 5 test cases

ID	Test case	Expected result	Actual result	Status
TC-067	Vendor current due = opening + Σ purchase due – account payments	$1,000 + 2,700 - 500 = 3,200$	3200	PASS
TC-068	Vendor statement totals	Opening 1,000; Purchase 6,200; Paid 4,000; Current 3,200	As expected	PASS
TC-069	Statement current due equals computeVendorDue (consistency)	Equal	As expected	PASS
TC-070	Account-level payments exclude per-purchase & rejected payments	500	As expected	PASS

ID	Test case	Expected result	Actual result	Status
TC-071	Vendor with no activity	Due 0	As expected	PASS

Factory Stock & Production — 7 test cases

ID	Test case	Expected result	Actual result	Status
TC-072	Current stock = loose + packed (kg)	$150 + 2 \times 15 = 180$	As expected	PASS
TC-073	All-time purchased / sold / other usage (production excluded)	350 / 210 / 5	As expected	PASS
TC-074	All-time opening derived so that Opening + In – Out = Closing	Opening 45, Closing 180	As expected	PASS
TC-075	September period stock (rolls back later movements)	Purchased 300, sold 195, usage 5, closing 145, opening 45	As expected	PASS
TC-076	Product ledger running balance ends at current stock	Balances 345,195,190,145,195,180	As expected	PASS
TC-077	Product ledger vendor summary	V1 300 kg first; total amount 14,100	As expected	PASS
TC-078	Production usage is not shown as stock leaving the factory	No ledger row for U2	As expected	PASS

Loans, Salary & Advances — 10 test cases

ID	Test case	Expected result	Actual result	Status
TC-079	Loan balance (rejected excluded, opening loan included)	Withdrawn 15,000; Repaid 5,000; Balance 10,000	As expected	PASS
TC-080	Monthly loan schedule spans first tx → later of last tx/today	Aug, Sep, Oct, Nov 2026	As expected	PASS
TC-081	Monthly schedule closing carries forward	Closings 5,000 / 11,000 / 11,000 / 10,000; last = balance	As expected	PASS
TC-082	Loan schedule carries across a year boundary	Jan 2027 opening 100, closing 60	As expected	PASS
TC-083	Loan transaction labels	Withdrawal / Repayment / Existing Loan / Balance Correction (+/–)	As expected	PASS
TC-084	Loan with no transactions	Empty schedule, balance 0	As expected	PASS
TC-085	Salary totals group names case/space-insensitively, skip rejected	Rahim: 11,000 over 2 payments, 2,000 advance adjusted	As expected	PASS
TC-086	Salary totals filtered by month	Oct 2026: 3,000 (1 payment)	As expected	PASS
TC-087	Advance salary tracker (given – adjusted)	Karim 1,000/0/1,000; Rahim 3,000/2,000/1,000	As expected	PASS
TC-088	Outstanding advance lookup & edit-exclusion	" RAHIM " → 1,000; excluding e3 → 3,000; blank → 0	As expected	PASS

Accounting (GL / TB / BS) — 7 test cases

ID	Test case	Expected result	Actual result	Status
TC-089	General ledger closing balances (incl. legacy "customer" alias)	Cash 1,200; AR 200; Sales –500; Rent 100	As expected	PASS
TC-090	General ledger ordered by account code with running balance	Codes ascending; Cash running 1,300 → 1,200	As expected	PASS
TC-091	Trial balance debits equal credits	Debit 1,500 = Credit 1,500	As expected	PASS
TC-092	Balance sheet balances with current-period profit in equity	Assets 1,400 = Liab 0 + Equity 1,000 + Profit 400	As expected	PASS
TC-093	Unmapped ledger entries are flagged, not crashing	1 unmapped (bank) entry	As expected	PASS
TC-094	Manual journal account resolves by accountRef	manual + c4 → Share Capital; no ref → undefined	As expected	PASS

ID	Test case	Expected result	Actual result	Status
TC-095	Accounting reports on null data	Empty, balanced	As expected	PASS

Dashboard — 4 test cases

ID	Test case	Expected result	Actual result	Status
TC-096	Dashboard counts & low-stock list	3 products, 2 dealers, 2 low-stock items	As expected	PASS
TC-097	Top dealers exclude rejected invoices	Dealer A 3,000 first; Dealer B 300	As expected	PASS
TC-098	Top products exclude rejected invoices	Chili 100g 3,000 first; Turmeric 300	As expected	PASS
TC-099	Recent invoices list (max 5, newest first)	First = INV-3	As expected	PASS

Users, Roles & Permissions — 5 test cases

ID	Test case	Expected result	Actual result	Status
TC-100	Permission ids generated from module × action matrix	49 unique ids	49	PASS
TC-101	Super Admin holds every permission	All ids granted	As expected	PASS
TC-102	Every default role grants only known permission ids	No unknown ids	16 roles checked	PASS
TC-103	hasPermission / getPermissions role checks	MD: orders:approve yes, users:create no; null user → false	As expected	PASS
TC-104	Permission definitions exist for every id	Labels for all ids	As expected	PASS

Configuration & Data Integrity — 11 test cases

ID	Test case	Expected result	Actual result	Status
TC-105	database.rules.json is valid JSON with erp rules	Parses; has rules.erp	46 rule nodes	PASS
TC-106	Every subscribed collection has a Firebase security rule	ERP_TOP_LEVEL_KEYS ⊂ rules.erp	45 keys covered	PASS
TC-107	Every approval collection is subscribed by the UI	APPROVAL_COLLECTIONS ⊂ ERP_TOP_LEVEL_KEYS	As expected	PASS
TC-108	Every sidebar link resolves to a page	All nav hrefs have page.tsx	19 routes	PASS
TC-109	Every expense category maps to a ledger account	No unmapped category	As expected	PASS
TC-110	Damage is not an Expense category (moved to Cash chart)	Damage ∉ EXPENSE_CATEGORIES; ∈ P&L cash categories	As expected	PASS
TC-111	Standard chart of accounts: unique codes & system accounts	No duplicate codes / ledgerAccounts	37 accounts	PASS
TC-112	Cash quick buttons & legacy renames point to real cash categories	All ∈ CASH_MAINTENANCE_CATEGORIES	As expected	PASS
TC-113	Books start date is a valid ISO date	2026-09-01	As expected	PASS
TC-114	Company letterhead constants populated	Name, address, email, helpline non-empty	Recipe Food Products Limited	PASS
TC-115	Starter catalog categories derived	Non-empty, unique	20 items / 5 categories	PASS

Build & Compile — 2 test cases

ID	Test case	Expected result	Actual result	Status
TC-116	TypeScript type check of whole project (tsc --noEmit)	0 type errors	0 errors (exit 0, 2.1 s)	PASS
TC-117	Next.js production build (next build --webpack)	Build succeeds; all admin routes generated	Compiled; 23 static pages generated (19 admin routes) plus 1 API route (exit 0, 31 s)	PASS